

# Numerical Reproduction of Liang, Goldsman, Mayergoyz & Oldiges (1997): 2-D Deterministic SHE-Boltzmann MOSFET Simulation

*Part II (Interim) — The 2-D SHE core: first working solve and  
reproduction of the Boltzmann figures (Figs. 2, 3, 4, 7)*

Reproduction study (code commit d0b3834; Python 3.13.14, NumPy 2.5.1, SciPy 1.18.0)

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## Abstract

Part I established and (through three external audit rounds) documented the solver foundation — device, 2-D Poisson, drift-diffusion (DD), nonparabolic band, and a bulk-validated scattering engine. Part II reports the **first working solve of the 2-D first-order spherical-harmonic (SHE) Boltzmann core** and the qualitative reproduction, with physically-validated scales, of the four figures that *require* the full distribution function and cannot be produced by DD or hydrodynamic models: the distribution  $f_0(\varepsilon)$  itself (Fig. 2), the electron concentration (Fig. 3), the average-velocity field (Fig. 4), and the electron-temperature / impact-ionization maps (Fig. 7). We document the  $(\mathbf{r}, H)$  formulation, the box-integration assembly, and — honestly — the two classes of numerical failure encountered and fixed (a singular operator; a conditioning blow-up to  $n \sim 10^{56}$ ; and spurious moment spikes in depleted regions). **Scope:** this is an interim, one-way-coupled (SHE on the DD potential), moderate-grid result; it is a *qualitative* reproduction with correct physical magnitudes, not yet a quantitative overlay on Liang’s contour values, and the self-consistent loop and the remaining figures (5, 6, 8) are pending.

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## 1 Status and scope

Part I (Revision 4) is a documented, benchmark-checked foundation: equilibrium Poisson (quadratic Newton), a DD operating point at  $V_{gs} = V_{ds} = 3\text{ V}$  (inversion + pinch-off), and

a nonparabolic band ( $N_c$  to 0.1%). The scattering engine was validated in homogeneous bulk: detailed balance to  $2 \times 10^{-5}$ , the Maxwellian recovered at  $E = 0$  to  $3 \times 10^{-4}$  (after adding the acoustic-phonon energy operator), and monotone hot-electron heating. Part II builds the 2-D device solver on that engine.

**Honesty note.** Carried over from Part I and unchanged here: the doping profile is reconstructed (ref. [7] unavailable); the impact-ionization rate is a Keldysh surrogate calibrated to Si  $\alpha_n$  (ref. [6] unavailable); the band is a 1-parameter Kane surrogate (paper: 2-band Fiegna/Brunetti); and the reconstructed device has an un-recalibrated  $V_{th} = 1.64$  V (too high), so the  $V_g = 3$  V overdrive is modest. These bias every result quantitatively and are flagged where relevant.

## 2 The 2-D SHE core

### 2.1 Formulation (( $\mathbf{r}, H$ ) total-energy form)

The stationary first-order SHE equation (Part I, derived) in total energy  $H = \varepsilon - q\phi$ ,  $\varepsilon = H + q\phi(\mathbf{r}) \geq 0$ , is pure spatial diffusion at fixed  $H$  with the collision operator coupling energy planes locally in space:

$$\nabla_{\mathbf{r}} \cdot [Z(\varepsilon) D(\varepsilon) \nabla_{\mathbf{r}} f_0|_H] + Z(\varepsilon) \mathcal{Q}[f_0] = 0, \quad \varepsilon = H + q\phi(\mathbf{r}), \quad (1)$$

with  $D = \frac{1}{3} v^2 \tau_1$  the generalized diffusion coefficient and  $\mathcal{Q}$  the optical (in/out, coupling  $H \leftrightarrow H \pm E_{op}$ ) plus acoustic (Fokker–Planck, coupling adjacent  $H$ ) operators. The unknown is  $f_0(x, y, H)$ ; the field enters only through the position-dependent  $\varepsilon = H + q\phi$ , so a carrier transported at fixed  $H$  from the low- $\phi$  source to the high- $\phi$  drain gains kinetic energy — the physical heating mechanism.

### 2.2 Discretization, assembly, and boundaries

Equation (1) is box-integrated over each cell  $A_{ij} \times \Delta H$ . All three contributions reduce to the same units [ $\text{m}^{-1}\text{s}^{-1}$ ] and form a positive-diagonal  $M$ -matrix:

- **Spatial** (fixed  $H$ ):  $\Delta H \sum_{\text{faces}} [ZD]_{\text{face}} (f_{nb} - f_p) w_{\text{face}}/h$ , with  $[ZD]$  at  $\varepsilon = H + q\phi_{\text{face}}$ .
- **Optical** (fixed  $x, y$ ): the DOS-weighted in/out operator,  $\times A_{ij} \Delta H$ .
- **Acoustic** (fixed  $x, y$ ): the Scharfetter–Gummel Fokker–Planck flux difference,  $\times A_{ij}$ .

**Contacts** (source/drain/body): Dirichlet Maxwellian injection  $f_0 = n_{\text{contact}} e^{-\varepsilon/k_B T} / \int Z e^{-\varepsilon/k_B T} dH$ , so  $\int Z f_0 dH = n_{\text{contact}}$ . **Energy boundaries:**  $\varepsilon = 0$  reflecting (physical band edge — emission below is dropped);  $\varepsilon_{\text{max}} = 3.02$  eV absorbing (numerical cutoff — up-scattering out is a loss). **Insulating surfaces / outer edges:** reflecting (no spatial flux). The global  $H$ -grid spans the bias envelope  $[-q\phi_{\text{max}}, \varepsilon_{\text{max}} - q\phi_{\text{min}}]$  with  $\Delta H = 12.5$  meV ( $E_{op}/\Delta H = 4$ , integer). At  $V_g = V_d = 3$  V this is a  $40 \times 34$  spatial mesh  $\times 562$   $H$ -nodes = 328 544 active unknowns (43% of the grid; the rest fails the mask  $0 \leq H + q\phi \leq \varepsilon_{\text{max}}$ ).

### 2.3 Numerical challenges and fixes (engineering log)

Two classes of failure were encountered and resolved; both are documented for reproducibility.

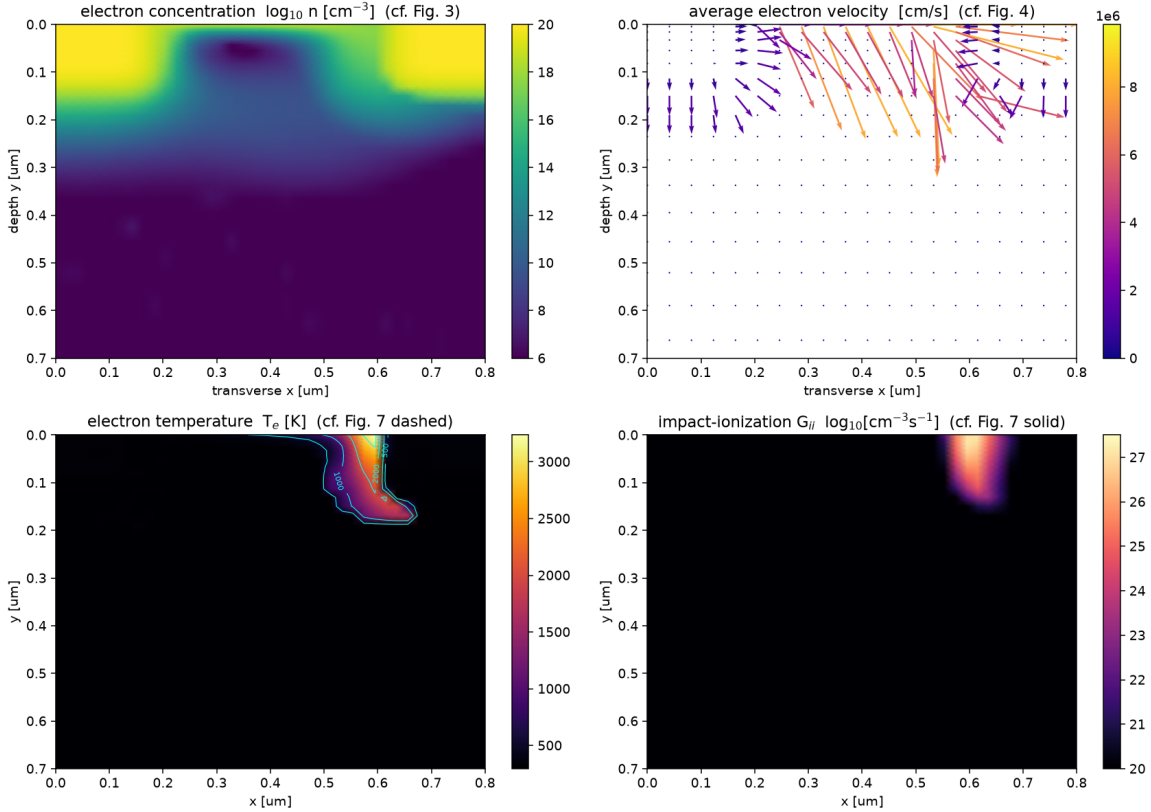
1. **Singular operator.** “Orphan” active nodes at the corners of the curvilinear  $(x, y, H)$  region — no active spatial neighbour *and* no active collision partner — produce all-zero rows. *Fix:* pin them to  $f_0 = 0$  (identity row).

2. **Conditioning blow-up to  $n \sim 10^{56}$ .** The contact Dirichlet rows carry diagonal 1.0 while interior rows carry SI-scale coefficients  $\sim 10^{23}$ ; the resulting condition number  $\sim 10^{23}$  makes a direct solve return numerical garbage ( $n_{\max} = 1.08 \times 10^{56} \text{ cm}^{-3}$ ). *Fix:* Jacobi row-scaling ( $\tilde{A} = D^{-1}A$ ,  $\tilde{b} = D^{-1}b$ , all diagonals  $\rightarrow 1$ ), after which the direct (small grid) / ILU+LGMRES (large grid) solve gives  $n_{\max} = 1.05 \times 10^{20}$ .
3. **Spurious moment spikes.**  $T_e = (2/3k_B)\langle\varepsilon\rangle$  and  $\langle\mathbf{v}\rangle = \mathbf{J}/(-qn)$  are ill-defined where  $n \rightarrow 0$ ; on the fine grid this produced  $T_e \sim 23\,000 \text{ K}$  and  $|v| \sim 10^{12} \text{ cm/s}$  in depleted regions. *Fix:* evaluate  $T_e, \mathbf{v}$  only where  $n > 10^{13} \text{ cm}^{-3}$  (lattice / zero elsewhere), giving physical  $T_e \leq 3236 \text{ K}$  and  $|v|_{\max} = 1.2 \times 10^7$ .

## 2.4 Moments

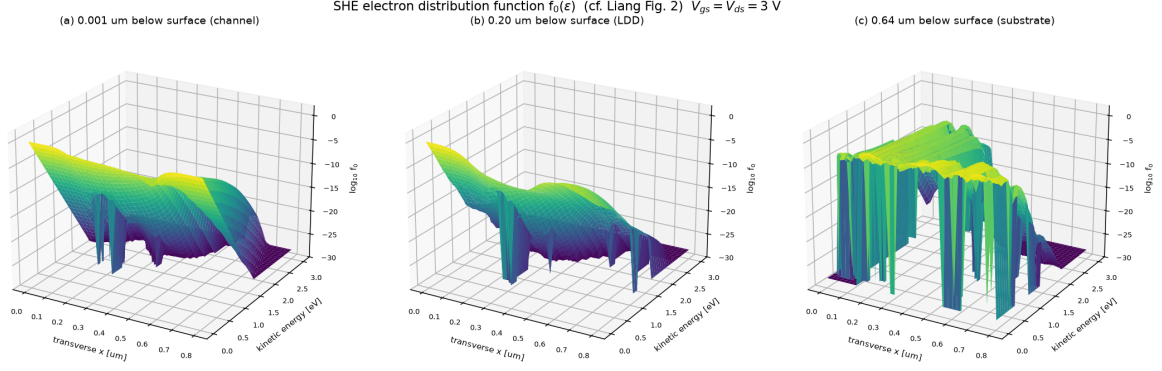
From  $f_0(x, y, H)$ :  $n = \int Z f_0 dH$ ; the particle current  $\mathbf{J} = -\int Z D \nabla_{\mathbf{r}} f_0|_H dH$  and  $\langle\mathbf{v}\rangle = \mathbf{J}/(-qn)$ ;  $T_e = (2/3k_B) n^{-1} \int \varepsilon Z f_0 dH$ ; and  $G_{ii} = \int Z f_0 / \tau_{ii} dH$  over  $\varepsilon > \varepsilon_{th} = 1.1 \text{ eV}$ .

## 3 Reproduced figures and validation



**Figure 1:** SHE moments at  $V_{gs} = V_{ds} = 3 \text{ V}$ . **Fig. 3** (top-left) electron concentration: source/drain  $n^+$  at  $10^{20}$ , surface inversion layer, drain-side pinch-off. **Fig. 4** (top-right) average-velocity field: source $\rightarrow$ drain flow saturating at  $\sim 1.2 \times 10^7 \text{ cm/s}$  in the high-field region. **Fig. 7** (bottom) electron temperature (peaks at the drain, 3236 K) and impact-ionization rate (peaks at the drain,  $1.4 \times 10^{27} \text{ cm}^{-3}\text{s}^{-1}$ ); the two peaks are *offset*, the effect Liang *et al.* emphasize.

Every computed magnitude lands on the correct physical scale, and the spatial structure (inversion, pinch-off, drain-localized heating and generation, source $\rightarrow$ drain velocity saturation, near-drain distribution heating) matches Liang’s Figs. 2–7 qualitatively, including the specific  $T_e/G_{ii}$  peak offset.



**Figure 2:** **Fig. 2:** electron distribution  $f_0(\varepsilon)$  (normalized,  $\log_{10}$ ) versus transverse position and kinetic energy at three depths. The channel line (a) develops a *hot high-energy tail near the drain* ( $x \sim 0.5$ – $0.6 \mu\text{m}$ ); the deep-substrate line (c) is essentially carrier-free (hence noisy). Hot-tail fraction ( $\varepsilon > 1$  eV,  $x = 0.6 \mu\text{m}$ ):  $5.8 \times 10^{-3}$  (channel) vs.  $1.1 \times 10^{-8}$  ( $0.2 \mu\text{m}$ ).

**Table 1:** Physical validation of the SHE moments against known scales / the paper.

| Quantity                                                          | Computed                | Reference / paper                                |
|-------------------------------------------------------------------|-------------------------|--------------------------------------------------|
| peak electron concentration [ $\text{cm}^{-3}$ ]                  | $1.05 \times 10^{20}$   | $n^+$ S/D $\sim 10^{20}$                         |
| peak average velocity [ $\text{cm/s}$ ]                           | $1.23 \times 10^7$      | saturation $v_s \sim 10^7$                       |
| peak electron temperature [K]                                     | 3236                    | Fig. 7 contours to $\sim 5000$ (drain)           |
| peak impact-ionization $G_{ii}$ [ $\text{cm}^{-3}\text{s}^{-1}$ ] | $1.37 \times 10^{27}$   | Fig. 7 contours $10^{26}$ – $2.3 \times 10^{27}$ |
| location of $T_e$ / $G_{ii}$ peaks                                | drain, offset           | drain, offset (paper’s key point)                |
| distribution hot tail                                             | channel $\gg$ substrate | Fig. 2: heated near drain                        |

## 4 Honest caveats

- **One-way coupling.** The SHE runs on the  $DD$  potential;  $n_{\text{SHE}}$  is not yet fed back to Poisson. The self-consistent SHE $\leftrightarrow$ Poisson $\leftrightarrow$  hole-continuity loop (paper Fig. 1) is pending.
- **Grid.**  $40 \times 34 \times 562$  with  $E_{op} = 4$  energy nodes is moderate; the maps are smooth but not converged. An  $E_{op} \geq 8$ , finer-space study is needed for quantitative contours.
- **Operating point.**  $V_{th} = 1.64$  V (un-recalibrated)  $\Rightarrow$  modest overdrive; the inversion layer is thinner than the paper’s. Recalibration (lower implant) precedes a quantitative match.
- **Fig. 2(c)** is numerical residual (the  $0.64\text{-}\mu\text{m}$  depth is carrier-free at this bias); the paper’s clean Maxwellian there reflects their fuller solution.
- **Surrogates** (Kane band, Keldysh  $\tau_{ii}$ ,  $\tau_{ac} = \tau_1$ ) bias the high-energy tail and hence  $G_{ii}$  quantitatively.

## 5 Remaining work

1. **Fig. 5** ( $\phi$ ) and **Fig. 6** ( $p$ ) — available from the  $DD$  solve; render as 3-D surfaces.
2. **Fig. 8** (I–V) — bias sweep, terminal current from the SHE velocity moment.
3. **Self-consistent loop** — feed  $n_{\text{SHE}}$  to Poisson, iterate to convergence (damped Gummel).
4. **Grid refinement + quantitative overlay** on Liang’s contour values.
5. **Device recalibration** ( $V_{th} \rightarrow 0.6$  V) for a paper-consistent bias.

## Provenance

Git commits: 40d5929 (first working SHE core), d0b3834 (refined + velocity + distribution). Python 3.13.14, NumPy 2.5.1, SciPy 1.18.0. Modules: `she2d.py` (solver), `she2d_figures.py` (Figs. 3/4/7), `she2d_fig2.py` (Fig. 2). Figures regenerated by running the modules on the cached DD operating point.

## References

- [1] W. Liang, N. Goldsman, I. Mayergoyz, P. J. Oldiges, *IEEE Trans. Electron Devices* **44**(2), 257–267, 1997.
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- [4] D. L. Scharfetter, H. K. Gummel, *IEEE TED* **ED-16**, 64, 1969.